5 Assembly and Dismantling RR3000 ISO (Typ 2701)

DANGER



Risk of injury

Improper handling of the packaged door can lead to injury.

⇨ Place palette on an even surface.

⇨ Do not remove palette before starting assembly.

5.1 Inspection of the delivery

The door is largely pre-assembled on delivery. The barrel with the rolled up curtain is located on a special transport- and installation rack. The columns are packed on a separate pallet. Drive, control unit and accessories are packed in a cardboard box.

- Check door for damage incurred during transit
- Check that the delivery is complete by comparing it with the order documentation.

5.2 Preparation for assembly

DANGER



Risk of injury

Incorrect assembly of the door can lead to injury.

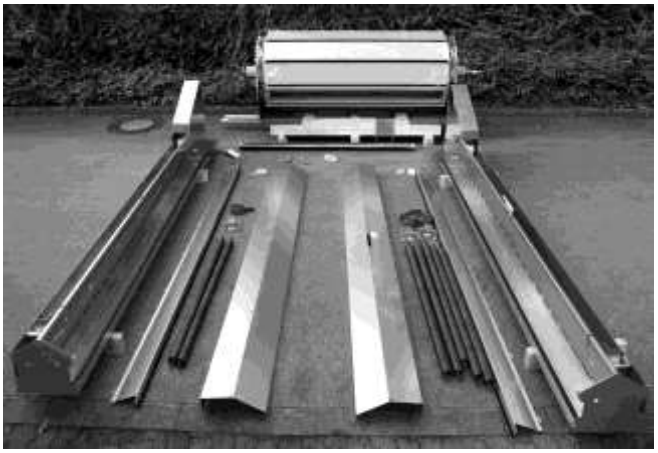
Isopanel walls, chequer bricks and light partition walls must be checked for their suitability for assembly.

⇨ The door may only be assembled by trained staff.

⇨ The door may only be assembled by the manufacturer or an installation firm approved by the manufacturer.

Make the following preparations before assembling the door:

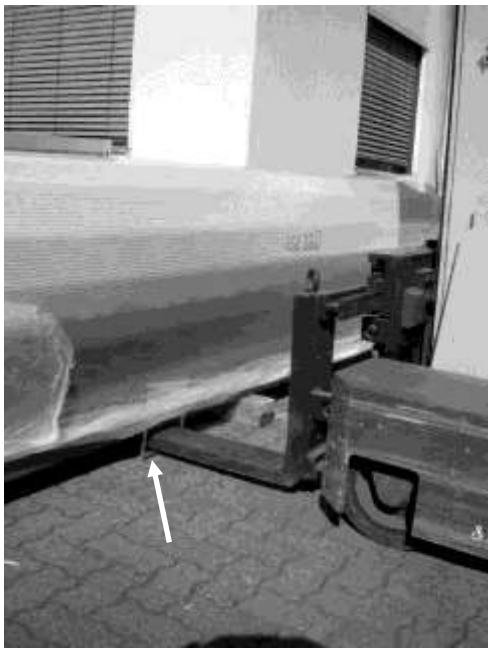
- Inspect the installation site for the necessary fixing points and assembly requirements: assemble a substructure or other suitable fixing options if necessary. Screw the upper fixing points to a steel substructure or concrete element. The fixing plan can be found in the user manual.
- Measure clearance dimensions (width and height) of the door opening and compare with the order.
- Cordon off the installation site against people and vehicle movements.
- Remove shrink-wrap from the packaged door.





1. The door shipment must be checked for transport damages and completeness – compare with delivery papers / order papers.

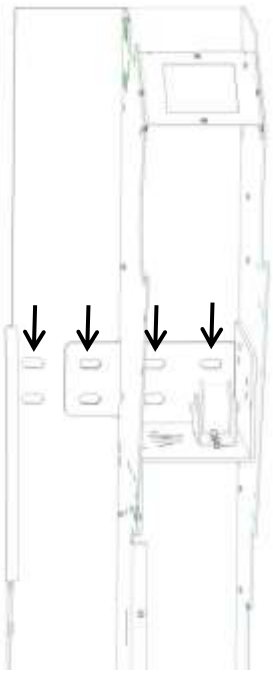
2. The drive and fixing material are delivered in a cardboard box.

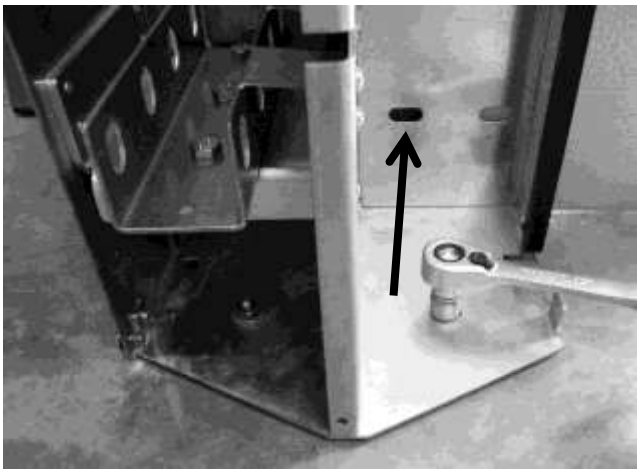




3. The barrel is placed on a pallet which works as a tool for transport and installation.
The transport-ring is to be used by the forklift driver as shown above to avoid toppling of the complete top section while transport.

4. If the floor declines the first side frame has to be arranged and fixed on the side of the higher level.
The first side frame must be placed exactly perpendicular.





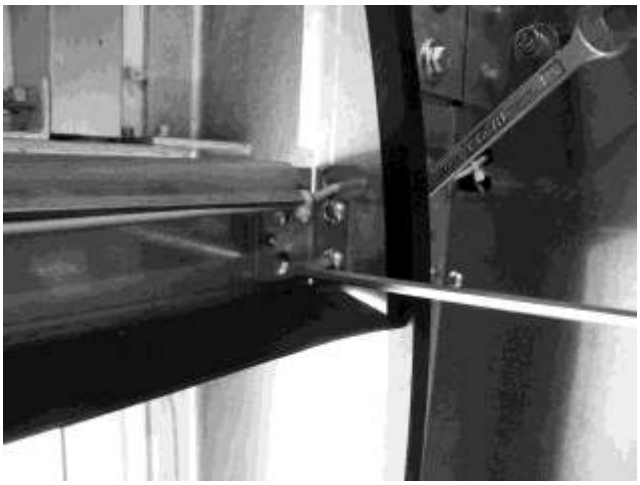
5. Fix the side frame within the upper area with at least 4 screws.

You can do this by temporarily removing the pulleys.

6. Fix the side frame within the lower area with at least 1 screw each chamber on the floor or alternatively on the wall.
For easier access you can reinstall the lower spring brackets temporary.

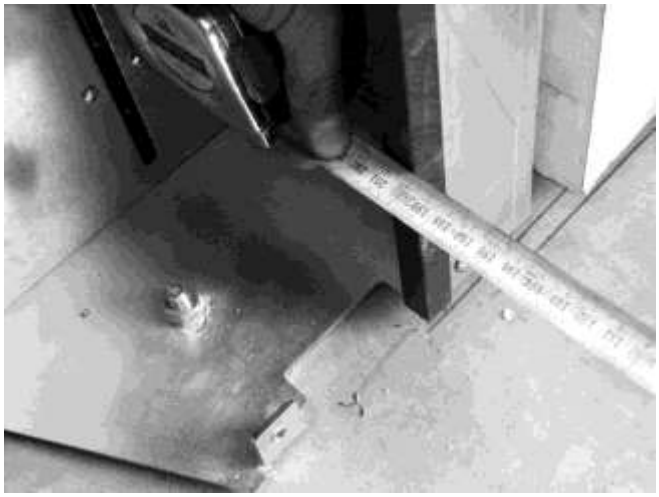
Set on the second side frame and place the bottom plate if necessary.

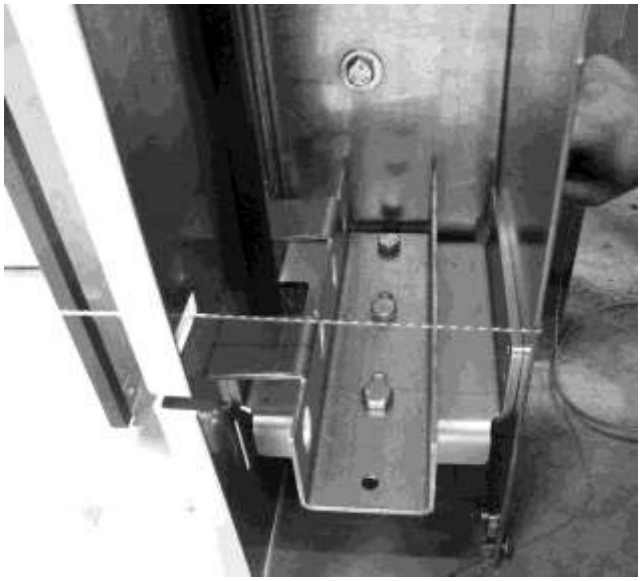
Fasten it up and down loosely on the wall or the floor.



7. Installation of second side frame

8. Fix the lintel profile within the upper area of the door with two screws and nuts M8 (installation bag) on the side frame.
Insert the connection cables on the lintel profile to the control system (fixing with cable binders).

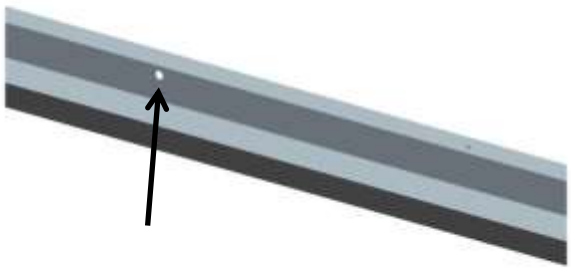


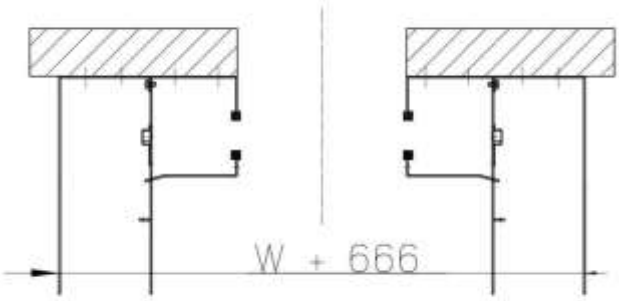


9. Measure down the clear door width (W) between the plastic rails.

10. Adjust a string to the side frames to align them.

Now screw the second side frame to the wall or floor.



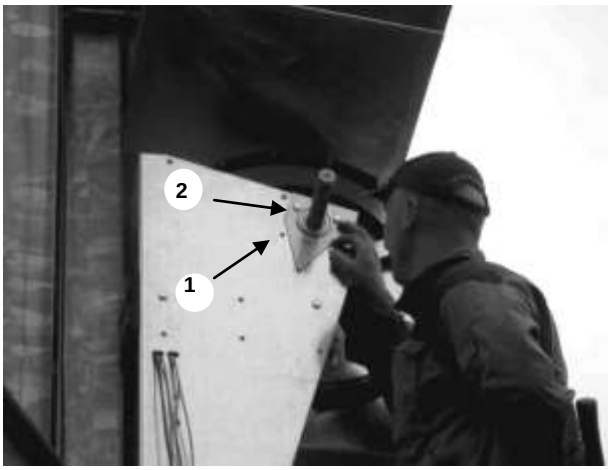


11. Fix the lintel profile for doors W>4000mm centrally at the wall.

12. Check up the upper distance between the side frames: W + 666mm.
Use distance plates next to the mounting screws to adjust the side frames.



Crushing hazard



With moving down the palette the barrel could rotate!

Do not remove the transport safety belts!

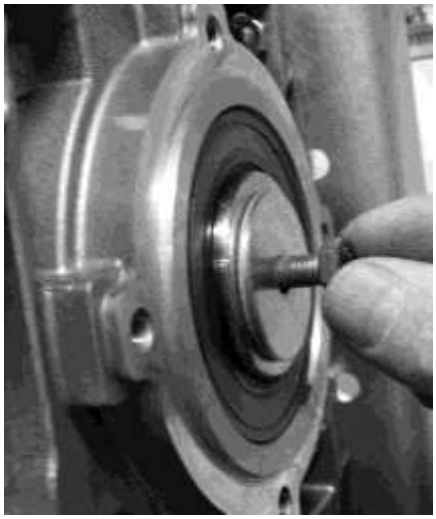
13. Lift up the barrel with assistance of a crane, forklift or equivalent device and lay down the complete barrel in the side frames as shown above.

14. Screw on the mounting plate with the side frame (1). Afterwards tighten the 3 screws of the flange-bearing (2).

Move down the palette carefully.

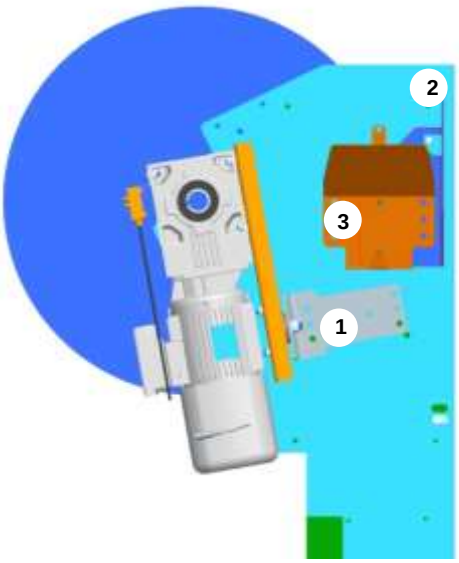




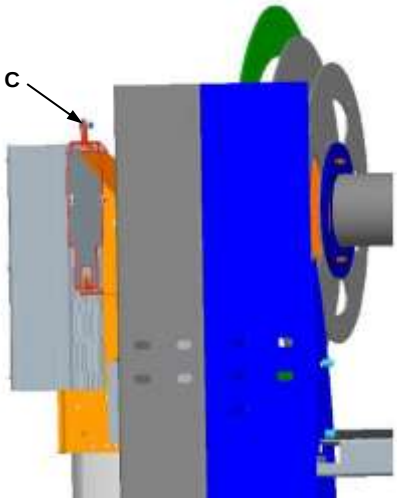


15. Turn the hollow shaft of gear box according position of shaft key in barrel. (Socket wrench SW6 on the motor). Lift the drive and the control together and slide the gear drive onto the shaft of the barrel.

16. Secure the drive to the drive shaft extension with a washer and locking screw M8x20 (installation bag).

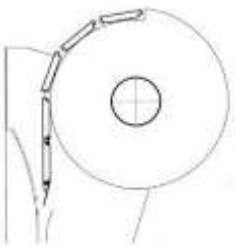


17. Connect the torque arrester (1), bracket motor cover (2) and bracket MCC (3) with the side frame.



18. Hang up control system (MCC) into bracket and fix position with screw (C).
Connect MCC according separate manual.

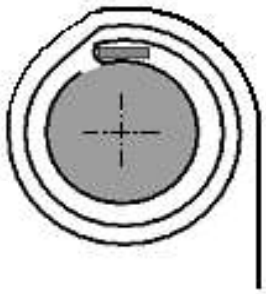




inside



outside

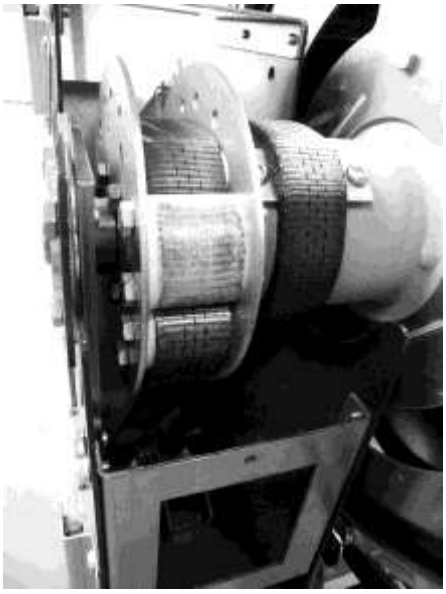


inside

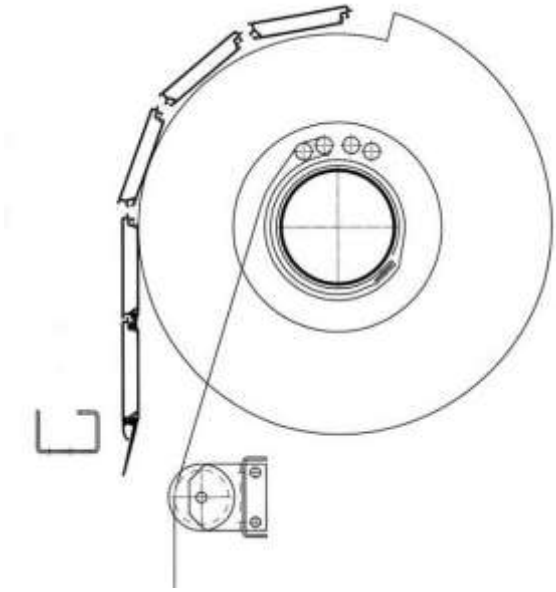
19. Turn the barrel into correct position and remove the transport belts.
(correct position: bottom lamella point downward).

20. Mount the belts for the counterbalance system (number and arrangement on the door see appendix).

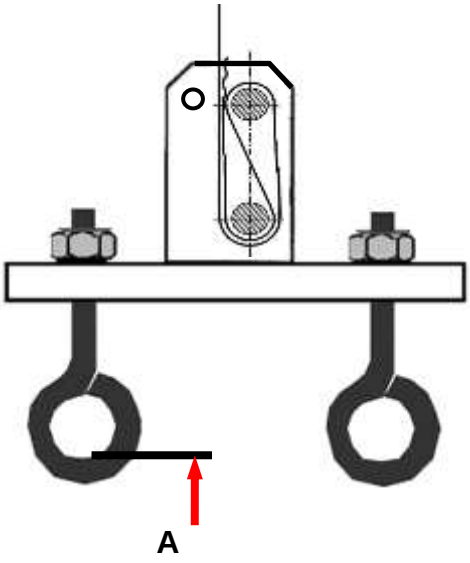
Place two dead windings over the clamp plate (door open).



21. For doors with more than 10 springs, the outer belt on the non-drive side is already pre-assembled. The belt is guided through a special device (AGS).



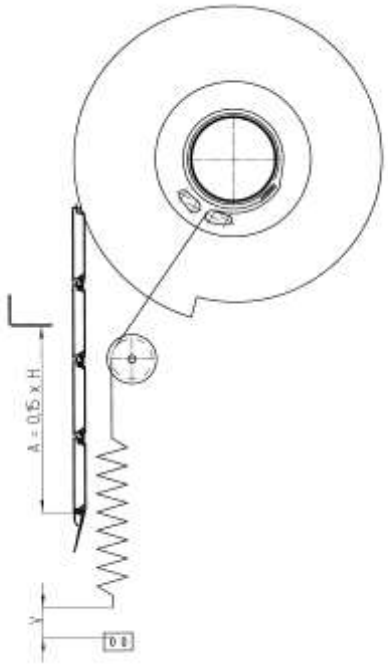
22. In this version (AGS) the outer belt is wound in the same direction as the curtain (when the door is open).



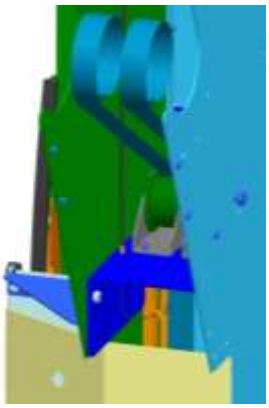
23. Fix the belts onto the upper spring acceptance according sketch.
(position see appendix)
Distance A between upper and lower spring bracket = length of spring + pre tension (V):
H<=5500mm: **V=250mm**
H>5500mm: **V=150mm**

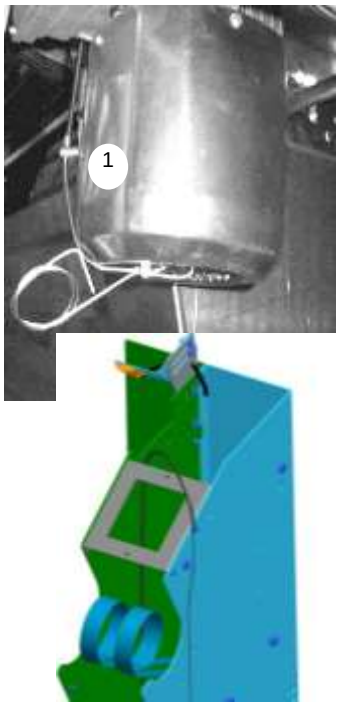


24. Hang in the springs first into the upper string acceptances and then into the lower acceptances of the side frames (alternately inside, outside form back to front).
Split the spring positions according appendix.

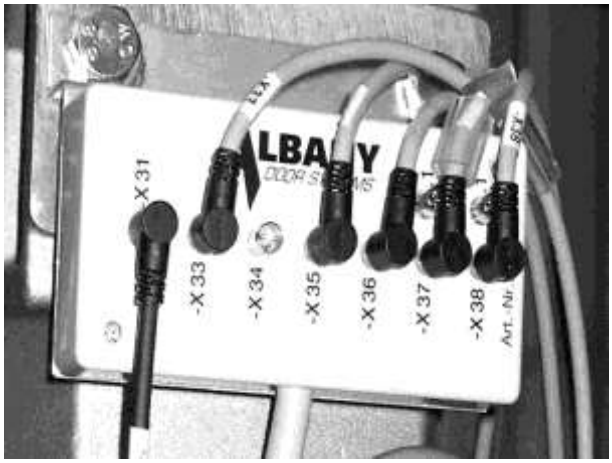


25. At doors with belt deflection system (AGS) turn the barrel into a neutral position (picture)
- door appr. 85% oben -
and hang in the spring with pre tension V.





26. Emergency handle: Pull the steal rope of the Bowden cable through the drills within the head plate and connect this end with the lever of the emergency handle. The other end of the cable must be able to affect the pin (1) of the brake as shown above.



The markings at the plug board must correspond with the ones of the cables.

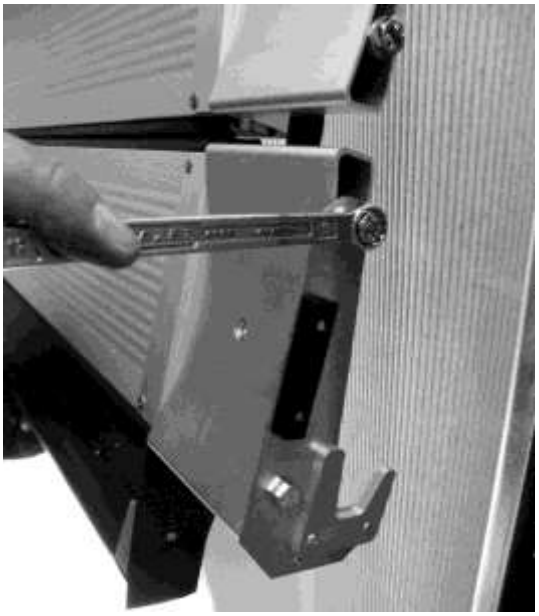


27. Connect the cables of the photocell or light grid to the plug board on the drive.

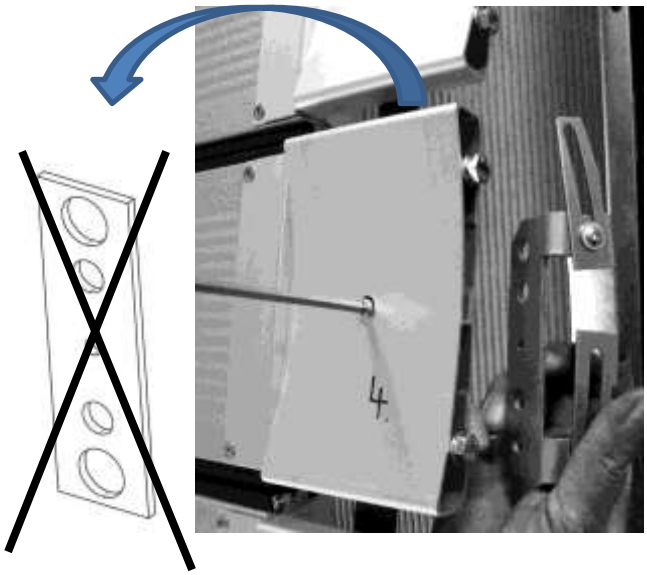
28. Install the display on drive side to the side frame (screw and nut M6 in installation bag) and connect it to the display cable of control system (MCC).



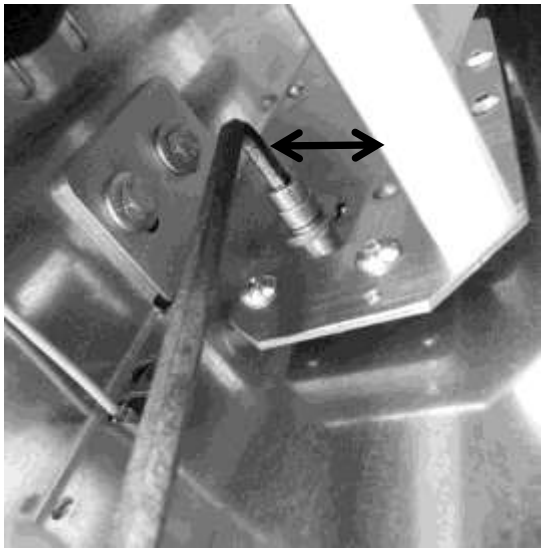
29. Connect earth connection from building (min. 6 mm²).



30. Close the door down to approx. 800 mm above the floor and fix the catching hooks on both sides of the bottom lamella.



31. Remove the clamping plate on the fourth lamella from the bottom (drive side) and attach the control flag with bracket.



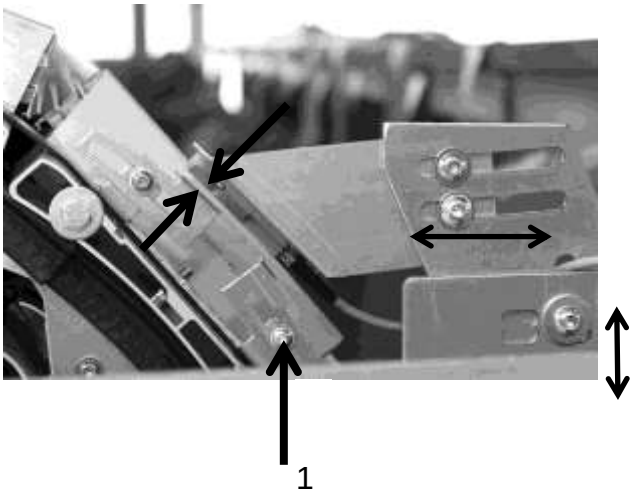
32. Open the door manually fully and adjust the stop bolts for the catching hook.



33. For doors with electrical safety edge (option) connect the cable chain with the bottom lamella.



34. Install the reference switch on the drive side and align these with a distance of 3mm to the flag (figure 37). Depending on the door height different positions are possible.



35. By moving the control flag (1) You can make a fine adjustment of the upper end position.



36. Attach the bottom guide covers first and finally the upper guide cover into the side frames.

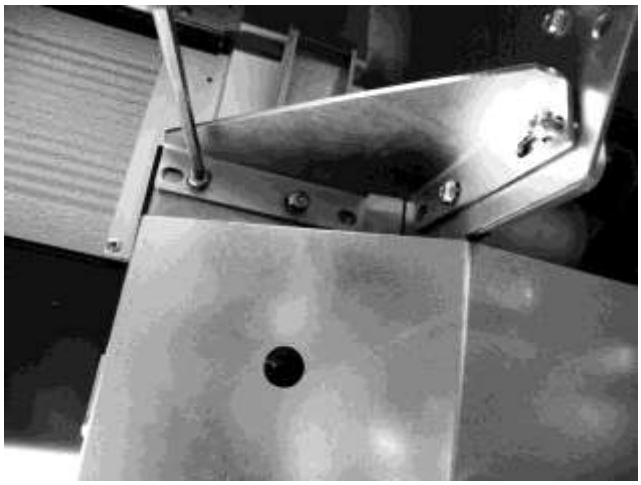
During disassembly of the lower cover, the upper cover can be unhooked and fall down.





37. Mount the side frame cover from bottom to top on both sides.
Install all screws only loosely.
(Flange bolts M8 x16 installation bag).

38. Close the door partially and press first the upper guide cover with plastic rails to the curtain. Now screw the side frame cover in the middle, then inside and last on the outside.
Now close the door completely and fix the bottom covers.

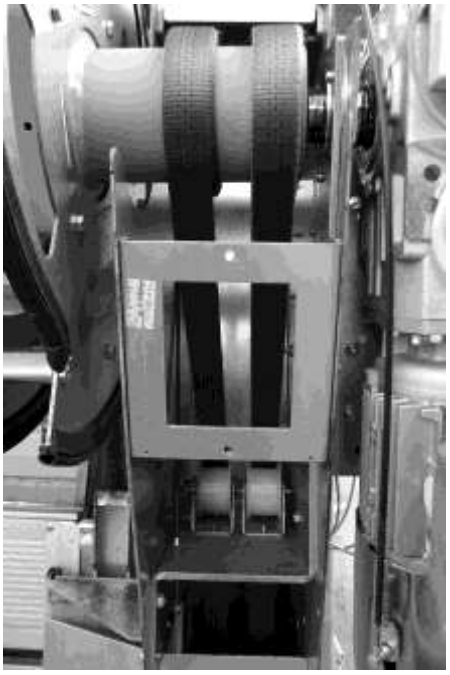
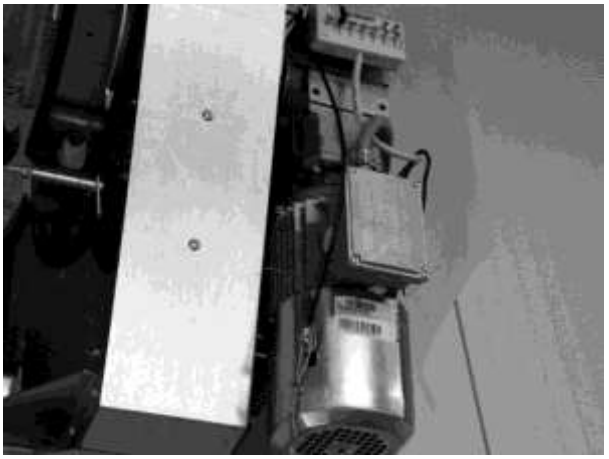


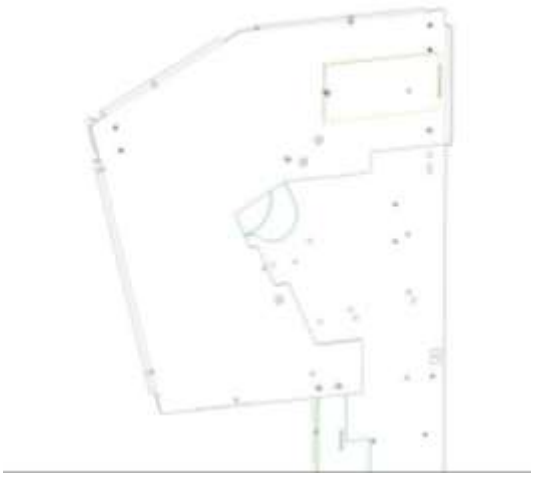


39. Screw the stiffeners at the upper end of the guide cover. Make sure that the gap is at least 56mm wide.

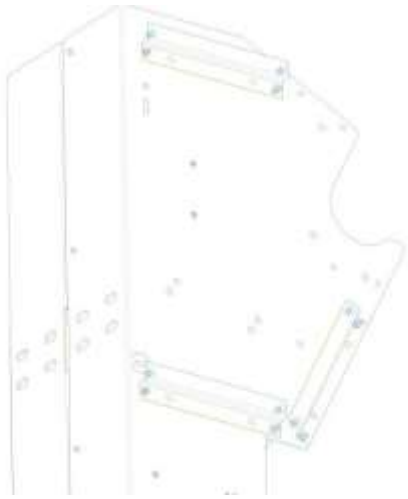
40. For doors H>3000 mm the guide cover is split in two or three parts.

Ensure a smooth transition to the black plastic rails. Make sure that the edges have a small chamfer.

	Door functions: - safeties - actuators - transition of the plastic rails - counterbalance system ⚠ At no position the doorblade may slide down to the floor
41. Connect the power supply to the control system and take them into operation (see separate instructions). Program the end positions and do some test runs.	**42.** Test the door functions. Pay attention to running noises.
	
43. The belts of the counter balance must exactly roll onto each other. By adjustment (move to right or left or twist) of the belt rollers, the winding behavior can be changed.	**44.** Install the top cover of the spring boxes. For doors with top roll cover this assembly is done later.



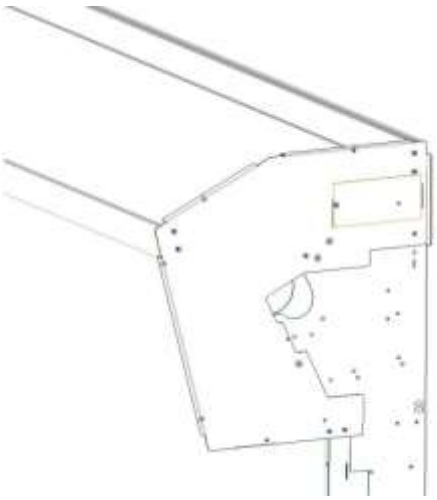
View: drive side rh, without motor



45. For doors with top roll cover you first install the side cover on the drive side with 5 screws M8x16 and 2 nuts (below).
For all other assembly steps the door is to close and disconnected from the power supply.

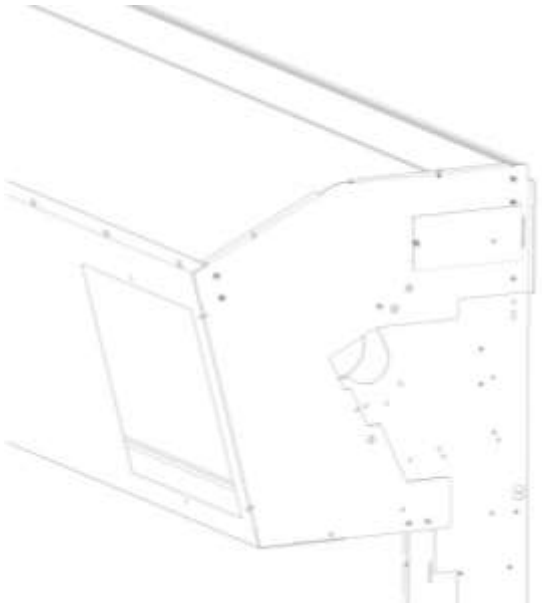
46. Now mount on the non-drive side the three Z-holder with 6 screws M8x16 and three nuts.



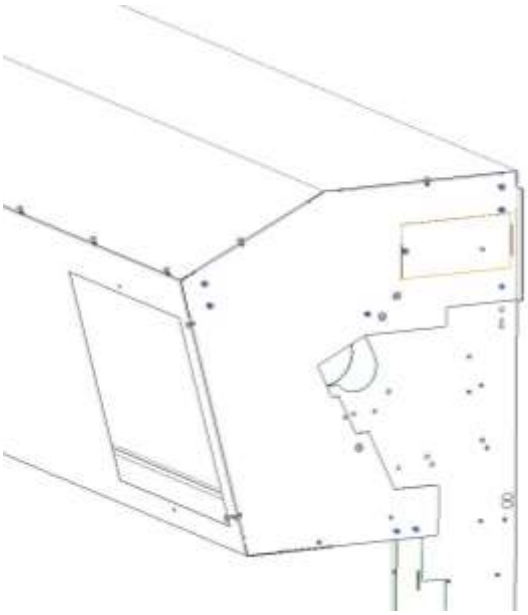


47. Fix the side frame cover on the non-drive side

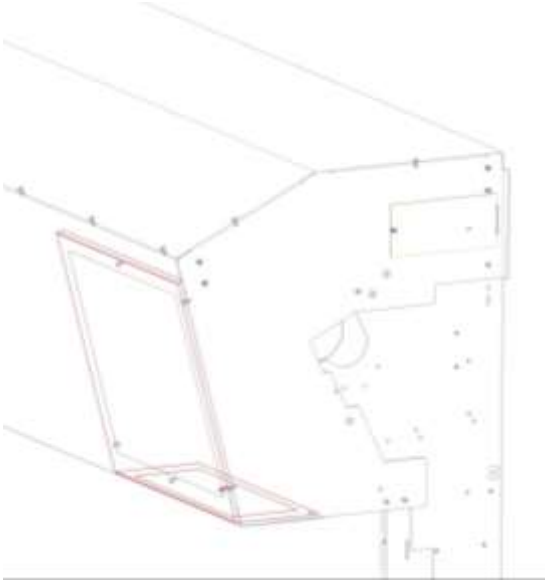
48. Mount the support lamellas first backwards, then forwards between the side frames.
The anterior lamella is covered with a black tape.



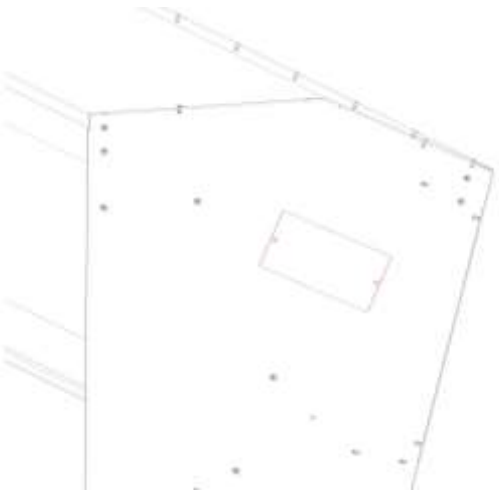
49. Attach the front cover and screw it to the side covers.
Remove protective film before.



50. Hang up the upper cover and fasten it with the side and the front covers.
Install the cover of the reference switch.

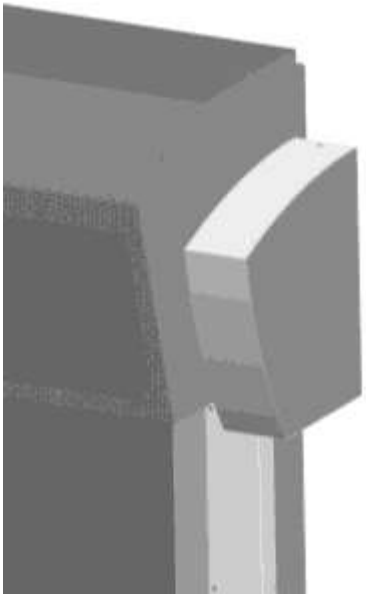


51. Install the front revision sliding covers.



52. At door version B: Mount the cover on the non-drive side.

The installation of the top roll cover is completed.

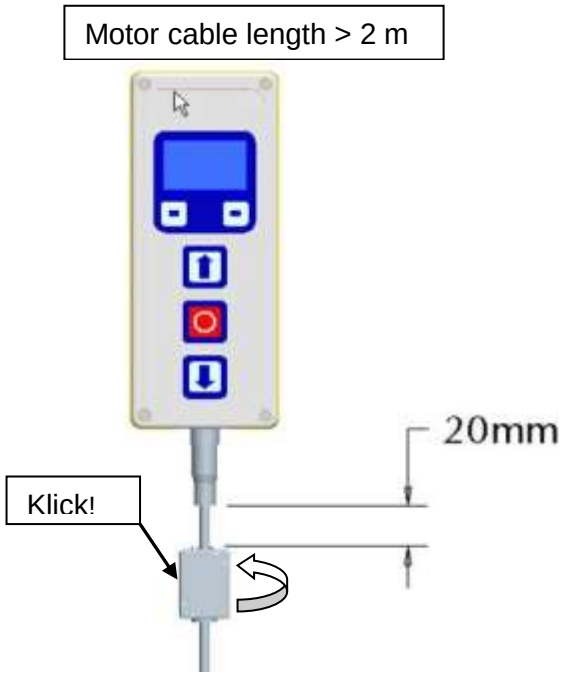


53. Install the motor cover (option).

Remove any protective films.

Install the spring box cover top (see 44).

54. Instruct the customer in the handling of the door.



After testing the safety equipment and a test run the installation is ready.

To dismantle, complete the steps backwards.

The door packaging can be returned to the producer.

Defect components should be disposed of onsite according to local laws.

Electronic components also need to be disposed of onsite according to local laws

55. If motor cable length (between drive and MCC) is longer than 2m install ferrite on display cable.

Gesamtzahl Feder	Seitenteil Antriebsseite Side frame drive side							Seitenteil Lagerseite Side frame non drive side							Door type
	Federaufnahme oben Spring bracket top			Federzahl No of springs			Anzahl Gurt No of belt	Federaufnahme oben Spring bracket top			Federzahl No of springs			Anzahl Gurt No of belt	
see 8020RG043	Pos. A	Pos. B	Pos. C	Pos. A	Pos. B	Pos. C	**	Pos. A	Pos. B	Pos. C	Pos. A	Pos. B	Pos. C	**	
4	3-Feder	3-Feder		2	2		2								5000A/B
5	3-Feder	3-Feder		3	2		2								5000A/B
6	3-Feder	3-Feder		2	2		2	3-Feder			2			1	5000A/B
7	3-Feder	3-Feder		3	2		2	3-Feder			2			1	5000A/B
8	3-Feder	3-Feder		3	2		2	3-Feder			3			1	5000A/B
9	3-Feder	3-Feder		3	3		2	3-Feder	3-Feder		3			1	5000A/B
10	3-Feder	3-Feder		3	2		2	3-Feder	3-Feder		3	2		2	5000A/B
11	3-Feder	3-Feder		3	3		2	3-Feder	3-Feder*		3	2*		2	5000A/B
12	3-Feder	3-Feder		3	3		2	3-Feder	3-Feder*		3	3*		2	5000A/B
13	3-Feder	3-Feder		3	3		2	3-Feder	4-Feder*		3	4*		2	5000A/B
14	3-Feder	4-Feder		3	4		2	3-Feder	4-Feder*		3	4*		2	5000A/B
15	4-Feder	4-Feder		4	4		2	3-Feder	4-Feder*		3	4*		2	5000A/B
16	4-Feder	4-Feder		4	4		2	4-Feder	4-Feder*		4	4*		2	5000A/B
17	5-Feder	4-Feder		5	4		2	4-Feder	4-Feder*		4	4*		2	5000B
18	5-Feder	4-Feder		5	4		2	5-Feder	4-Feder*		5	4*		2	5000B
19	5-Feder	5-Feder		5	5		2	5-Feder	4-Feder*		5	4*		2	5000B
20	5-Feder	5-Feder		5	5		2	5-Feder	5-Feder*		5	5*		2	5000B
21	5-Feder	5-Feder		5	5		2	5-Feder	5-Feder*	3-Feder	4	5*	2	3	5000C
22	5-Feder	5-Feder		5	5		2	5-Feder	5-Feder*	3-Feder	5	5*	2	3	5000C
23	5-Feder	5-Feder		5	5		2	5-Feder	5-Feder*	3-Feder	5	5*	3	3	5000C
24	5-Feder	5-Feder		5	5		2	5-Feder	5-Feder*	4-Feder	5	5*	4	3	5000C
25	5-Feder	5-Feder		5	5		2	5-Feder	5-Feder	5-Feder	5	5*	5	3	5000C

* mit asynchronen Gewichtsausgleich
* with asynchronous balancing system

Leeres Feld = "0"
empty field = "0"

Gurt / belt **	Item-No.
5000A	5105R0045
5000B, C	5105R0100

Federaufnahme Spring bracket	Item-No.
3-Feder	8220R0050
4-Feder	8220R0051
5-Feder	8220R0058

C B A

Draufansicht Seitenteil links / rechts gespiegelt
Top view side frame left / right mirror-inverted